Programming Assignment #2

Floorplanning with Symmetry Constraint

Due: May 3, 2026

1 Problem Description

Let $B = \{b_1, b_2, b_3, \ldots, b_n\}$ be a set of n rectangular hard IP modules/blocks which can be rotated. The floorplanning with symmetry constraints problem is to place all modules within a rectangular chip without any overlaps such that the area of chip bounding box, A is minimized while the symmetry constraint is satisfied. We assume that the lower-left corner of this chip is at the origin (0,0) and no space (channel) is needed between two modules. The symmetry constraint is defined as follows: Let $S = \{S_1, S2, \ldots, S_k\}$ be a set of k symmetry groups, and each symmetry group has its own symmetry axis. A symmetry group $Si = \{(b_1, b'_1), (b_2, b'_2), \ldots, (b_p, b'_p), b^s_1, b^s_2, \ldots, b^s_q\}$ consists of p symmetry pairs and q self-symmetry modules, where (b_j, b'_j) denotes a symmetry pair and b^s_k denotes a self-symmetry module. A symmetry pair, containing two modules with the same dimensions and orientations, should be placed symmetrically along the symmetry axis. A self-symmetry module must have its center point placed on the symmetry axis.

2 Input

The input file starts with the total number of hard IP blocks, including symmetry and non-symmetry blocks, followed by the description of n blocks, and all symmetry constraints. The description of each block contains the block name, followed by the width and height of the block. Each symmetry constraint contains a number of symmetry pairs and self-symmetry blocks. The input format and a sample input are given as follows:

Input Format	Sample Input
NumBlock: # // Number hard IP blocks	NumBlocks: 9
<Module Name> <Module Width> <Module Height>	b1 100 50
⋮	b2 50 100
	b3 50 100
	b4 75 75
	b5 45 75
	b6 120 75
	b7 100 40
	b8 100 40
	b9 40 70
	b10 60 70
Symmetry Group	Symmetry Group
<Module Name> <Module Name> // Symmetry Pair	b2 b3
⋮	b1
<Module Name> // Self-symmetry module	Symmetry Group
⋮	b7 b8
	b9
	b10

3 Output

In the program output, you are asked to give the bottom-left coordinate of each block. The following table gives the output format and an output to the sample input.

Output Format	Sample Output
<Block Name> <x> <y>	b1 0 0
⋮	b2 0 50
	b3 50 50
	b4 100 0
	b5 75 0
	b6 100 75
	b7 220 0
	b8 220 110
	b9 220 40
	b10 260 40

4 Language/Platform

1. Language: C or C++ is preferred.
2. Platform: Linux.

5 Command-line Parameter

In order to test your program, you are asked to add the following command-line parameters to your program (e.g., Floorplan input.dat output.dat):

[executable file name] [input file name] [output file name]

6 Submission

You need to submit a tar file, which includes (1) source codes, (2) Makefile, (3) a text readme file (readme.txt) stating how to build and use your program, to E3 (`https://e3.nycu.edu.tw/`) by the deadline. Please put all required files in a folder, and use the following command to compress the folder in the linux environment.

tar cvf Student_ID.tar Student_ID

The folder name and the compressed file name must be your student ID.

7 Grading Policy

This programming assignment will be graded based on (1) the **correctness**, (2) **solution quality**, and (3) **running time**. For each test case, it will be regarded as "failed" if your program takes more than 1 hours.

There will be 20% penalty per day for late submission.